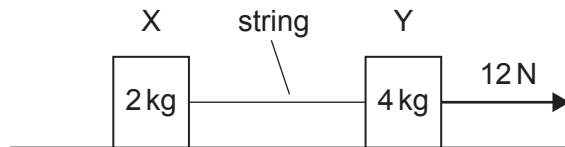


## PAPER 1 / QUESTION 48

4 November 2021 / TZ0 / HL Paper 1 / original question 3 / 8821-6501

48. X and Y are two objects on a frictionless table connected by a string. The mass of X is 2 kg and the mass of Y is 4 kg. The mass of the string is negligible. A constant horizontal force of 12 N acts on Y.



What are the acceleration of Y and the magnitude of the tension in the string?

|    | Acceleration of Y/ $\text{ms}^{-2}$ | Tension in the string/N |
|----|-------------------------------------|-------------------------|
| A. | 2                                   | 4                       |
| B. | 2                                   | 6                       |
| C. | 3                                   | 4                       |
| D. | 3                                   | 6                       |